# Ashritha Gonuguntla

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Software and ML systems engineer, 3 yrs at Cisco (Senior). C++/CUDA inference, LLM fine-tuning and serving, distributed telemetry.

## Education

**Carnegie Mellon University, School of Computer Science** — Pittsburgh, PA

Master of Software Engineering, **GPA 4.0/4.0** — Aug 2025 – Dec 2026

- Relevant Coursework: Deep Learning Systems, Generative AI, Computer Systems, AI Agents. TA: ML in Production, Computer Systems.
- **Honors:** Mary Shaw Award for Creativity in Software Engineering '26 · 5th place, Gray Swan AI Red-Teaming Challenge '26 · CMU Moon Miners, NASA Lunabotics '26 · JN Tata & KC Mahindra Scholar '25 · Smart India Hackathon Winner '20

**K L University**, B.Tech Computer Science, AI & Robotic Process Automation, GPA 9.84/10, rank 2 — Jun 2018 – Mar 2022

## Experience

**eParts Services (CMU MSE Studio client), Technical Lead, ML & Architecture** — 2026 – Present

- Leading **5 engineers** shipping a confidence-routed attribute-prediction service to the client: BGE+FAISS retrieval, a 198K-pattern rule engine, and Mahalanobis confidence routing over 4.08M attribute rows; 95.74% accuracy on 143K held-out predictions.

**Cisco, Software Engineer → Senior Software Engineer** — Aug 2022 – Jul 2025

- Tech-led 9 engineers across Cisco Secure Firewall's two telemetry paths in C++/Go: a gRPC plane with control/data channels, protobuf metrics, mTLS, ACK-based cancellation, plus gNMI/OpenConfig YANG subscriptions; scale-tested to **1,000 collector endpoints**.
- Contributed to **LLM post-training** for a 1B security model, later released as Antares-1B: prepared CVE/CWE SFT data, evaluated early checkpoints, built network-disabled Docker sandboxes for read-only repo exploration; scope preceded GRPO.
- Rebuilt the intrusion-event stream-processing and query path with tailored indexing and range-based partitioning: **40% lower latency at 2x telemetry volume**; mentored 4 engineers through design and performance reviews.
- Cut median SOC triage time **57%** with an agentic RAG incident assistant: multi-step LLM orchestration and tool calling over firewall telemetry and configuration, OpenSearch vector retrieval on AWS SageMaker/Lambda, root-cause summaries.

**Cisco, Software Intern** — Jan 2022 – Jun 2022

- Built cloud-native security observability on Kubernetes with **Prometheus, Grafana, CloudWatch**; automated AWS provisioning with Terraform, Ansible, CloudFormation.

**Adobe Research, Research Intern** — May 2021 – Aug 2021

- Co-authored Outage-Watch (ACM ESEC/FSE'23), a mixture density network for QoS tail-risk forecasting: **AUC 0.98, MTTD cut up to 88%**; co-inventor on US Patent App. 17/656,263.

**Microsoft, Software Engineering Intern** — May 2020 – Aug 2020

## Selected Publications

- Published *The Replay Gap: Static Evaluation of Model Switching in LLM Agents* (sole author, COLM'26 Workshop) · *The Plan, Not the Decoder* (sole author, ECCV'26 Workshop Oral) · *Multi-LoRA Composition via Softmax Routing* (AAAI'27, under review).

## Selected Projects

**nanoinfer** | *C++20, ARM NEON, CUDA, cuBLAS/cuDNN, ONNX Runtime* — Code

- Built a C++20 edge inference engine (im2col+SGEMM, NEON depthwise, fused epilogues, liveness-planned arena with no forward-pass allocations): **0.99x ONNX Runtime** on the dense CNN, 0.60–0.77x on depthwise, 37–77% less activation memory.
- Wrote an int8 CUDA GEMM ladder (dp4a, WMMA, smem-staged 32x32 warp tiles) hitting **122.8 TOPS**, ahead of cuBLAS to n=1024 and 0.71x at 4096; diagnosed the first WMMA kernel bandwidth-bound (8.6 GB over a 1008 GB/s bus) and staged it 4.6x faster.

**batch-invariance** | *vLLM 0.28, Qwen3-1.7B, RTX 4090, CUDA graphs* — Code

- Built a probe harness isolating batch size, batch composition, repeat rate, and throughput against vLLM 0.28, and priced the determinism fix at **54 to 67%** of throughput, a cost the documentation states but never quantifies.

**specdec** | *PyTorch, Qwen3-0.6B draft, Qwen3-4B target, RTX 4090* — Code

- Implemented speculative decoding from scratch and showed it **loses at every window size** in eager PyTorch (0.87x to 0.51x): the draft costs 0.649 of a target pass, so break-even reduces to acceptance = cost ratio.

**Query-Rewrite Evaluation** | *Qwen3-1.7B LoRA, BM25 from scratch, QReCC, nDCG, Kendall $\tau$* — Code

- Fine-tuned Qwen3-1.7B with LoRA on 48,415 QReCC pairs (0.400 to **0.481 nDCG@10** vs. a 0.505 human reference) and showed ROUGE-L ranks 7 rewriting systems backwards from retrieval (Kendall $\tau$ $-0.73$ to $-0.87$).

## Technical Skills

**Languages:** Python, C++20, Go, Java, SQL, Bash, TypeScript.
**ML and LLMs:** PyTorch, Hugging Face, vLLM, LoRA/PEFT, SFT data curation, quantization (FP8, AWQ, int8), speculative decoding, causal tracing, RAG, tool calling, MCP, LangChain.
**Systems and GPU:** CUDA (dp4a, WMMA), cuBLAS, cuDNN, ONNX Runtime, ARM NEON, gRPC, Protocol Buffers, Kafka, Linux, profiling, indexing and partitioning.
**Retrieval and Eval:** BM25, BGE, FAISS, OpenSearch, nDCG, Kendall $\tau$, paired bootstrap, calibration, agent benchmarks, fault injection.
**Cloud and Delivery:** Docker, Kubernetes, AWS, GCP, Terraform, Ansible, Prometheus, Grafana, GitHub Actions, FastAPI, pytest.